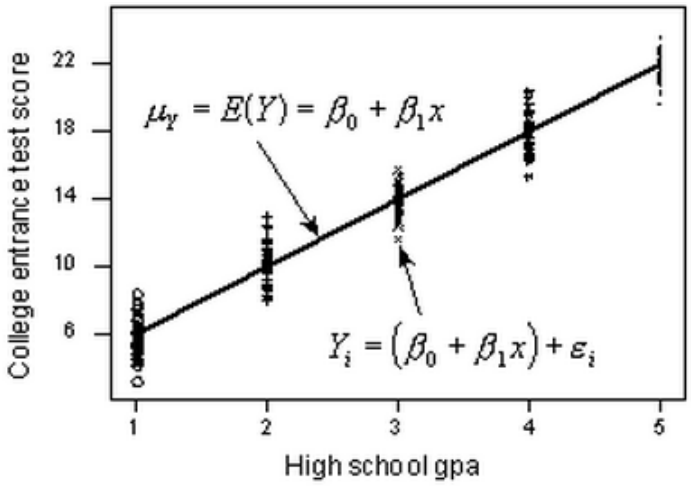
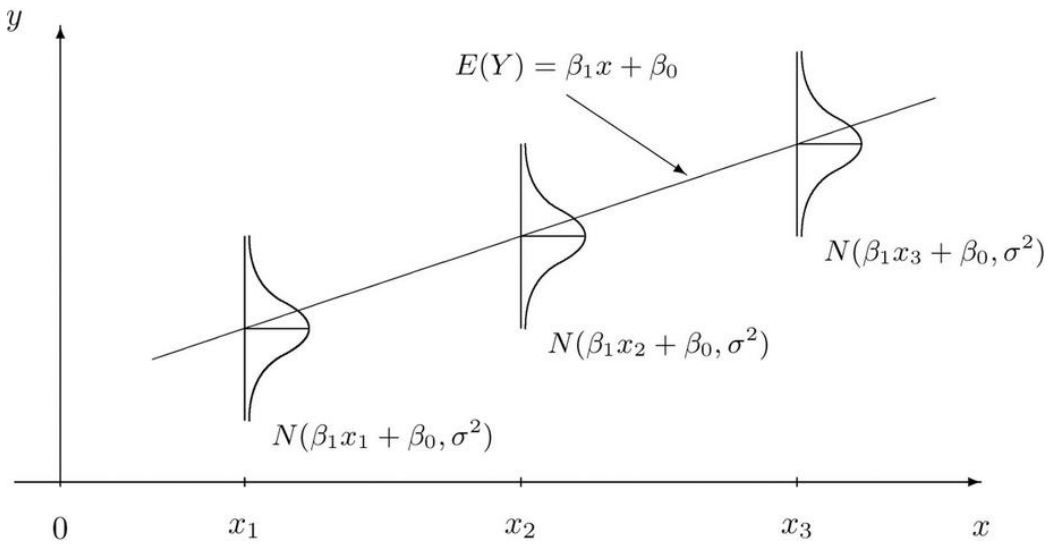
Chapter 5: Simple Linear Regression





Fitting the model

The fitted line is given by $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$.

Let $S(\alpha, \beta) = \sum_{i=1}^{n} e_i^2$

$= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

$= \sum_{i=1}^{n} (y_i - (\hat{\alpha} + \hat{\beta} x_i))^2$

$\frac{\partial S}{\partial \alpha} = 0 \Rightarrow -2 \sum_{i=1}^{n} (y_i - (\hat{\alpha} + \hat{\beta} x_i)) = 0$

$\sum_{i=1}^{n} y_i = n\hat{\alpha} + \hat{\beta} \sum_{i=1}^{n} x_i$

$\hat{\alpha} = \frac{\sum_{i=1}^{n} y_i}{n} - \hat{\beta} \frac{\sum_{i=1}^{n} x_i}{n}$

$\therefore \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$ —①

$\frac{\partial S}{\partial \beta} = 0 \Rightarrow -2 \sum_{i=1}^{n} (y_i - (\hat{\alpha} + \hat{\beta} x_i))(x_i) = 0$

$\sum_{i=1}^{n} x_i y_i - \hat{\alpha} \sum_{i=1}^{n} x_i - \hat{\beta} \sum_{i=1}^{n} x_i^2 = 0$

$\sum_{i=1}^{n} x_i y_i = \hat{\alpha} \sum_{i=1}^{n} x_i + \hat{\beta} \sum_{i=1}^{n} x_i^2$ —②

①→②:

$\sum_{i=1}^{n} x_i y_i = \left(\frac{\sum_{i=1}^{n} y_i}{n} - \hat{\beta} \frac{\sum_{i=1}^{n} x_i}{n} \right) \sum_{i=1}^{n} x_i + \hat{\beta} \sum_{i=1}^{n} x_i^2$

$= \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} - \hat{\beta} \left(\frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} - \sum_{i=1}^{n} x_i^2 \right)$

$\therefore \hat{\beta} = \frac{\sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} = \frac{S_{xy}}{S_{xx}}$

Checking for minimum:

$\frac{\partial^2 S}{\partial \alpha^2} = 2n > 0$

$\frac{\partial^2 S}{\partial \beta^2} = 2 \sum_{i=1}^{n} x_i^2 > 0$

Thus $\hat{\alpha}$ and $\hat{\beta}$ are minimum estimators.

1) Show that $\sum_i^n e_i^2 = S_{YY} - \frac{S_{XY}^2}{S_{XX}}$.

Proof:

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad , \hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$$

$$= \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta} x_i)^2 \quad , \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

$$= \sum_{i=1}^n (y_i - (\bar{y} - \hat{\beta}\bar{x}) - \hat{\beta} x_i)^2$$

$$= \sum_{i=1}^n [(y_i - \bar{y}) - \hat{\beta}(x_i - \bar{x})]^2$$

$$= \sum_{i=1}^n (y_i - \bar{y})^2 - 2\hat{\beta}\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) + \hat{\beta}^2 \sum_{i=1}^n (x_i - \bar{x})^2 \quad , \hat{\beta} = \frac{S_{XY}}{S_{XX}}$$

$$= S_{YY} - 2\left(\frac{S_{XY}}{S_{XX}}\right)(S_{XY}) + \left(\frac{S_{XY}}{S_{XX}}\right)^2 (S_{XX})$$

$$= S_{YY} - \frac{S_{XY}^2}{S_{XX}} \quad \text{(shown)}$$

Thus $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^n e_i^2$

$$= \frac{1}{n-2}\left(S_{YY} - \frac{S_{XY}^2}{S_{XX}}\right) \#$$

2) Show that $E(\hat{\beta}) = \beta$ and $Var(\hat{\beta}) = \frac{\sigma^2}{S_{XX}}$.

$$\hat{\beta} = \frac{S_{XY}}{S_{XX}} = \frac{\sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

$$= \frac{\sum x_i y_i - \bar{x}\sum y_i}{\sum x_i^2 - n\bar{x}^2} \quad , \bar{x} = \frac{\sum x_i}{n}$$

$$= \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum x_i^2 - n\bar{x}^2}$$

$$E(\hat{\beta}) = E\left[\frac{\sum (x_i - \bar{x}) Y_i}{\sum x_i^2 - n\bar{x}^2}\right]$$

$$= E\left[\frac{\sum (x_i - \bar{x})(\alpha + \beta x_i)}{\sum x_i^2 - n\bar{x}^2}\right]$$

$$= \alpha E\left[\frac{\sum (x_i - \bar{x})}{\sum x_i^2 - n\bar{x}^2}\right] + \beta E\left[\frac{\sum (x_i - \bar{x}) x_i}{\sum x_i^2 - n\bar{x}^2}\right] \quad , E[\sum (x_i - \bar{x})] = E[\sum x_i - n\bar{x}] = E[n\bar{x} - n\bar{x}] = 0$$

$$= \beta E\left[\frac{\sum x_i^2 - \bar{x}\sum x_i}{\sum x_i^2 - n\bar{x}^2}\right] \quad , \sum x_i = n\bar{x}$$

$$= \beta$$

$$Var(\hat{\beta}) = Var\left(\frac{\sum(x_i-\bar{x})Y_i}{\sum x_i^2 - n\bar{x}^2}\right)$$

$$= \sum_{i=1}^{n} Var\left(\frac{(x_i-\bar{x})Y_i}{\sum x_i^2 - n\bar{x}^2}\right)$$

$$= \sum_{i=1}^{n} \frac{(x_i-\bar{x})^2}{(\sum x_i^2 - n\bar{x}^2)^2}\sigma^2 \quad , \because Var(Y_i)=\sigma^2 \ , \ \sum_{i=1}^{n}(x_i-\bar{x})^2 = \sum x_i^2 - 2\bar{x}\sum x_i^2 - n\bar{x}^2$$

$$= \frac{\sum x_i^2 - n\bar{x}^2}{(\sum x_i^2 - n\bar{x}^2)^2}\sigma^2 \qquad = \sum x_i^2 - 2\bar{x}(n\bar{x}) - n\bar{x}^2$$

$$= \frac{\sigma^2}{S_{xx}} \quad , \because S_{xx} = \sum x_i^2 - \frac{\sum x_i^2}{n} = \sum x_i^2 - n\bar{x}^2 \qquad = \sum x_i^2 - n\bar{x}^2$$

3) Show that $E(\hat{\alpha}) = \alpha$ and $Var(\hat{\alpha}) = \sigma^2\left(\frac{1}{n} + \frac{\bar{X}^2}{S_{XX}}\right)$.

proof:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

$$E(\hat{\alpha}) = E\left(\frac{\sum Y_i}{n} - \hat{\beta}_1\frac{\sum x_i}{n}\right)$$

$$= \frac{1}{n}\sum_{i=1}^{n}E(Y_i) - \frac{1}{n}\beta\sum_{i=1}^{n}x_i$$

$$= \frac{1}{n}\sum_{i=1}^{n}(\alpha+\beta x_i) - \frac{\beta}{n}\sum_{i=1}^{n}x_i$$

$$= \frac{1}{n}(n\alpha) + \frac{\beta}{n}\sum_{i=1}^{n}x_i - \frac{\beta}{n}\sum_{i=1}^{n}x_i$$

$$= \alpha$$

$$Var(\hat{\alpha}) = Var\left(\frac{\sum Y_i}{n} - \hat{\beta}\frac{\sum x_i}{n}\right)$$

$$= \frac{1}{n^2}\left[\sum_{i=1}^{n}Var(Y_i) + (\sum x_i)^2 Var(\beta)\right] \quad , Var(Y_i)=\sigma^2, \ Var(\beta)=\frac{\sigma^2}{S_{xx}}, \ \sum x_i = n\bar{x}$$

$$= \frac{1}{n^2}\left[n\sigma^2 + (n\bar{x})^2\left(\frac{\sigma^2}{S_{xx}}\right)\right]$$

$$= \frac{\sigma^2}{n} + \frac{\sigma^2\bar{x}^2}{S_{xx}}$$

$$= \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) \quad \text{(shown)} \ \#$$

4) Show that the MLE for σ^2 is $\left(\frac{n-2}{n}\right)\hat{\sigma}^2$, where $\hat{\sigma}^2$ is the LS estimator for σ^2.

(Linear regression). *Our data is n observations with one explanatory variable and one response variable. The model is that*

$$y_i = \alpha + \beta x_i + \epsilon_i$$

where the ϵ_i are independent mean 0 normal random variable. The (unknown) variance is σ^2. The likelihood function

$$\mathbf{L}(\alpha, \beta, \sigma^2 | \mathbf{y}, \mathbf{x}) = \frac{1}{\sqrt{(2\pi\sigma^2)^n}} \exp -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2.$$

$$\ln \mathbf{L}(\alpha, \beta, \sigma^2 | \mathbf{y}, \mathbf{x}) = -\frac{n}{2} \ln 2\pi\sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2.$$

This, the maximum likelihood estimators $\hat{\alpha}$ and $\hat{\beta}$ also the least square estimator. The predicted value for the response variable

$$\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i \quad , \quad \hat{\beta} = \frac{S_{xy}}{S_{xx}} \ , \ \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \ , \ \hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2$$

The maximum likelihood estimator for σ^2 is

$$\hat{\sigma}^2_{MLE} = \frac{1}{n} \sum_{k=1}^{n} (y_i - \hat{y}_i)^2 = \frac{1}{n}\sum_{i=1}^{n} e_i^2 = \frac{n-2}{n}\hat{\sigma}^2$$

Proof:

Using MLE method:

$$\frac{\partial L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} - \frac{(-1)}{2(\sigma^2)^2}\sum_{i=1}^{n}(y_i - (\alpha + \beta x_i))^2$$

$$\frac{\partial L}{\partial \sigma^2} = 0 \Rightarrow \frac{1}{2(\hat{\sigma}^2)^2}\sum_{i=1}^{n}(y_i - (\hat{\alpha} + \hat{\beta} x_i))^2 = \frac{n}{2\hat{\sigma}^2}$$

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - (\hat{\alpha} + \hat{\beta} x_i))^2$$

$$= \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$= \frac{1}{n}\sum_{i=1}^{n} e_i^2 \quad , \ \hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2 \text{ (From LS method)}.$$

$$= \left(\frac{n-2}{n}\right)\hat{\sigma}^2 \quad \text{(shown)} \ \#$$